

University of Utah

Spring 2026

# **MATH 3210-004**

## **Midterm 2 Questions**

Instructor: Alp Uzman

March 20, 2026, 8:35 AM - 9:25 AM

Surname:

First Name:

uNID:

KEY

**Before turning the page  
make sure to read and  
sign the exam policy  
document, distributed  
separately.**

1. [20 points] **Prove or disprove the following statement:** Let  $I$  be an interval,  $f_n : \mathbb{Z}_{\geq 0} \rightarrow F(I; \mathbb{R})$  be a sequence of real valued functions on  $I$  and  $f_\infty \in F(I; \mathbb{R})$  be arbitrary. If  $f_n$  converges to  $f_\infty$  uniformly, and  $f_n$  is bounded for all but finitely many  $n \in \mathbb{Z}_{\geq 0}$ , then  $f_\infty$  is bounded.

$$\begin{aligned} x, y \in I, \quad n \in \mathbb{Z}_{\geq 0} \Rightarrow & |f_\infty(x) - f_\infty(y)| \leq |f_\infty(x) - f_n(x)| \\ & + |f_n(x) - f_n(y)| + |f_n(y) - f_\infty(y)| \\ & \leq 2 d_C(f_n, f_\infty) + \text{diam}(f_n). \end{aligned}$$

$$\Rightarrow \boxed{\forall n \in \mathbb{Z}_{\geq 0} : \text{diam}(f_\infty) \leq 2 d_C(f_n, f_\infty) + \text{diam}(f_n)}$$

$$\exists N, \forall n \geq N : d_C(f_n, f_\infty) < 1$$

by unif. conv.

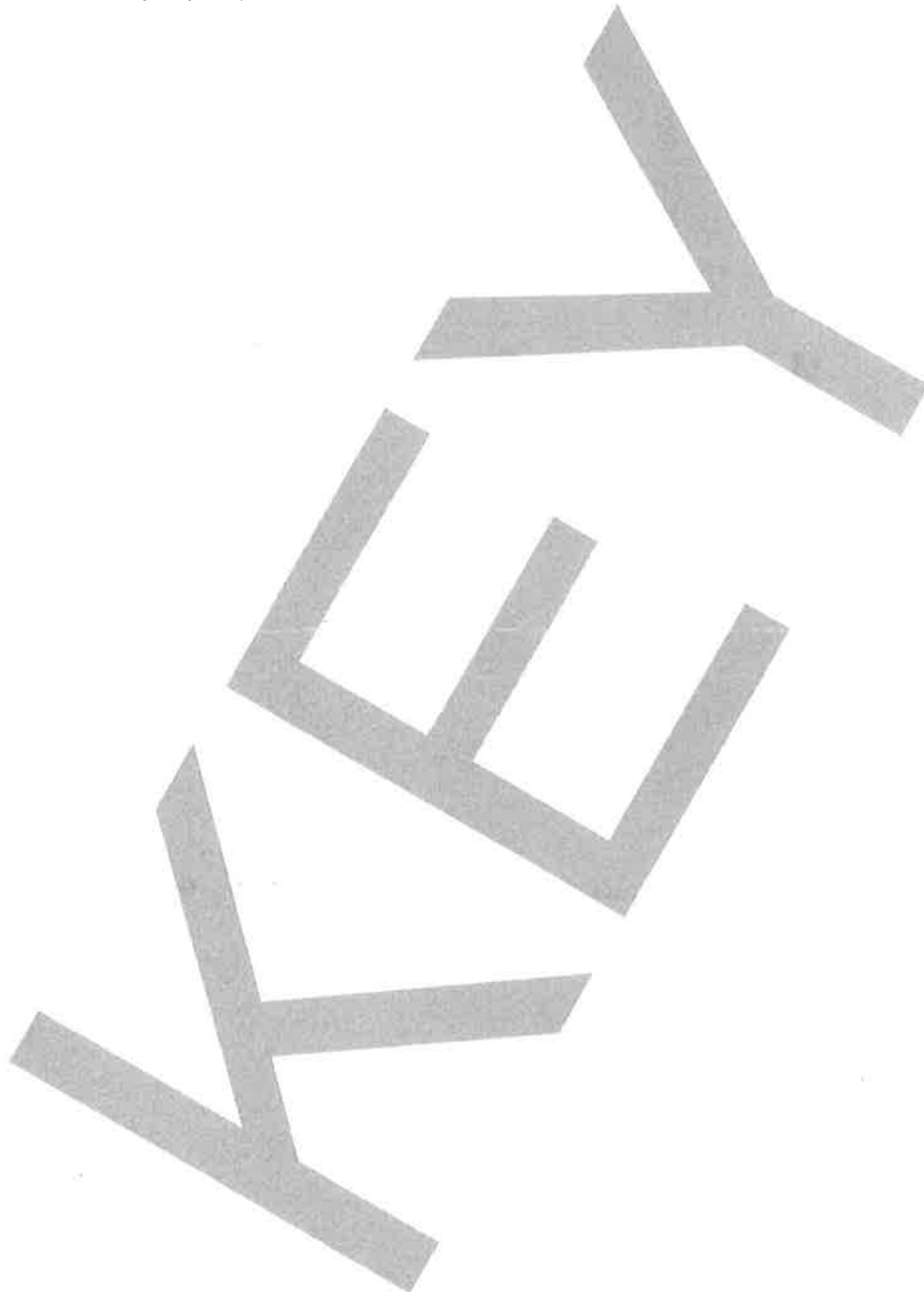
$$N_* = \max \{ n \mid f_n \text{ is unbounded} \} + 1 + N$$

finite set

Then for  $n = N_*$ ,

$$\text{diam}(f_\infty) \leq 2 + \underbrace{\text{diam}(f_{N_*})}_{< \infty} < \infty$$

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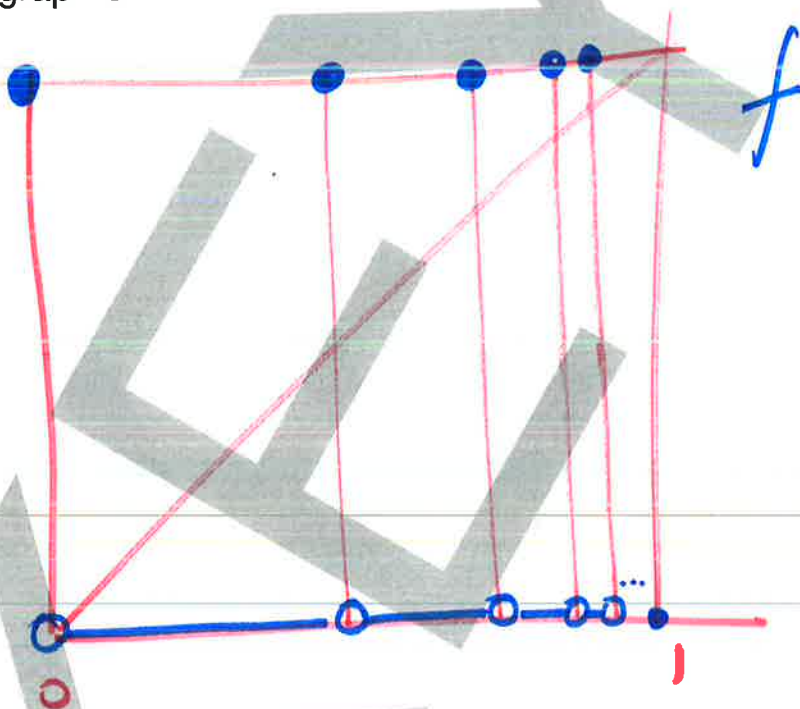


2. [63 points] Let  $I = [0, 1]$  and consider the function  $f: I \rightarrow \mathbb{R}$  defined by

$$f: x \mapsto \begin{cases} 1, & \text{if } x = 1 - \frac{1}{2^n} \text{ for some } n \in \mathbb{Z}_{\geq 0} \\ 0, & \text{else} \end{cases}$$

$$S_\alpha = \left\{ 1 - \frac{1}{2^n} \mid n \in \mathbb{Z}_{\geq 0} \right\}$$

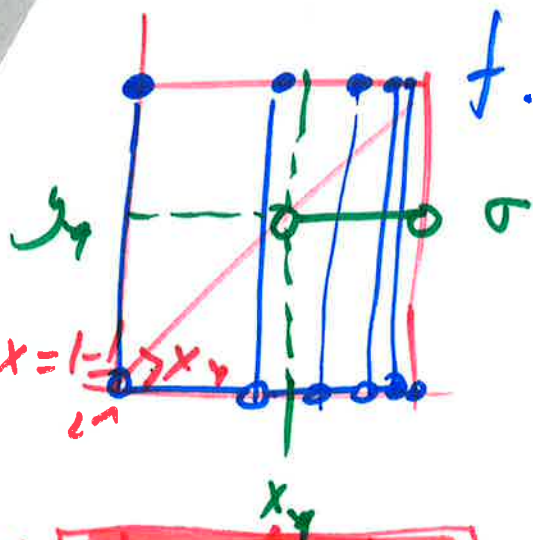
- (a) [40 points] There is nothing to prove in this part. Sketch the graph of  $f$ .



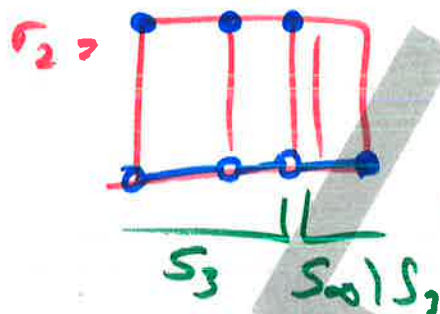
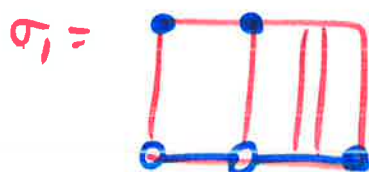
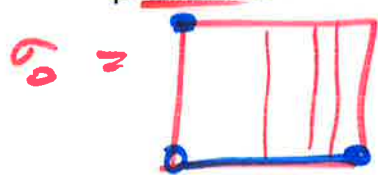


(d) <sup>3</sup> [3 points]Prove or disprove the following statement:There is a sequence of step functions on  $I$  that converges uniformly to  $f$ .Let  $\sigma : I \rightarrow \mathbb{R}$  be a step function.Then  $\sigma$  is continuous except at finitely many points.Define  $x_* = \max \{x \in [0, 1] \mid \sigma \text{ discontinuous at } x\}$ .Then  $\exists! y_* \in \mathbb{R}, \forall x \in ]x_*, 1[$ ,  $\sigma(x) = y_*$ .Thus if  $x \in ]x_*, 1[$ ,

$$|\sigma(x) - f(x)| = \begin{cases} |y_* - 1|, & \text{if } x = 1 \\ |y_*|, & \text{else.} \end{cases}$$



- (b)  points] Prove or disprove the following statement:  
There is a sequence of step functions on  $I$  that converges pointwise to  $f$ .



$$\forall n: \sigma_n: I \rightarrow \mathbb{R}, \quad \sigma_n(x) = \begin{cases} 1, & \text{if } x \in \left[1 - \frac{1}{2^n}, 1\right] \\ 0, & \text{else.} \end{cases}$$

then  $\forall n: \sigma_n$  is a step,  
if  $x \in I$ ,

$$|\sigma_n(x) - f(x)| = \begin{cases} 0, & \text{if } x \in S_n \\ 1, & \text{if } x \in S_{\infty} \setminus S_n \\ 0, & \text{else.} \end{cases}$$

But also, if  $1 - \frac{1}{2^n} > x$ ,

$$x \notin S_{\infty} \Rightarrow |\sigma_n(x) - f(x)| = 0.$$

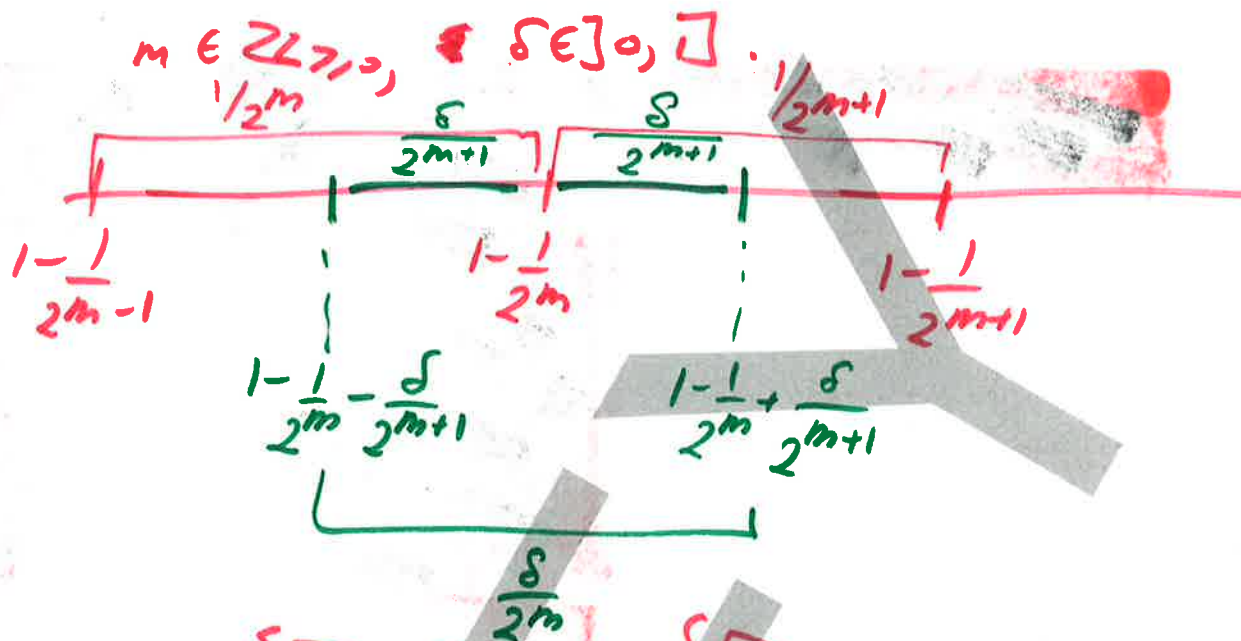
$$S_{\infty} = \left\{1 - \frac{1}{2^n} \mid n \geq 0\right\}$$

Since  $1 - \frac{1}{2^n} \uparrow 1$ ,  $\sigma_n \rightarrow f$  pointwise  
□



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(C) [10 points] Prove or disprove the following statement:

 $f$  is Riemann integrable and its Riemann integral is 0.

$$P_{m, \delta} = \left\{ \left[ 0, \frac{\delta}{4} \right] \right\} \cup \left\{ \left[ 1 - \frac{1}{2^m} - \frac{\delta}{2^{m+1}}, 1 - \frac{1}{2^m} + \frac{\delta}{2^{m+1}} \right] \right\}$$

$$1 \leq m \leq n$$

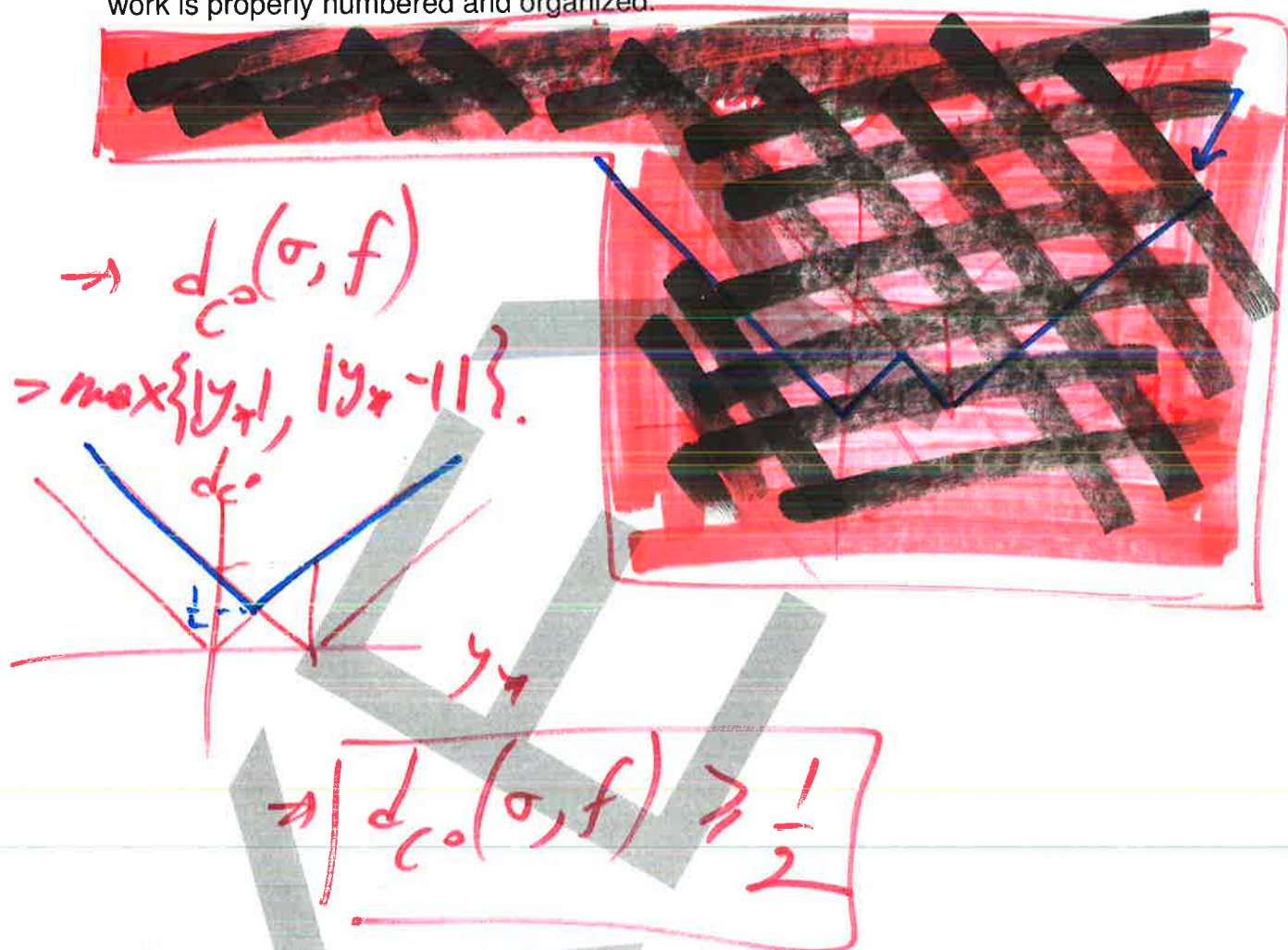
$$\cup \left\{ \left[ 1 - \frac{1}{2^n} + \frac{\delta}{2^{n+1}}, 1 \right] \right\} \cup \{ \text{Rest} \}$$

$$D(f; P_{m, \delta}) = U(f; P_{m, \delta}) = \frac{\delta}{4} + \sum_{m=1}^n \frac{\delta}{2^m} + \frac{1}{2^n} - \frac{\delta}{2^{n+1}}$$

$$= \delta \left( \frac{1}{4} + \sum_{m=1}^n \frac{1}{2^m} - \frac{1}{2^{n+1}} \right) + \frac{1}{2^n} \xrightarrow[\delta \rightarrow 0]{m \rightarrow \infty} 0$$

$$0 \leq \int_I f \leq U(f; P_{m, \delta}) \Rightarrow \boxed{\int_I f = 0} \Rightarrow \boxed{f \in R} \quad \square$$

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There are no step functions that  
are  $\frac{1}{2}$  close to  $f$  in  $C^0$   
metric.

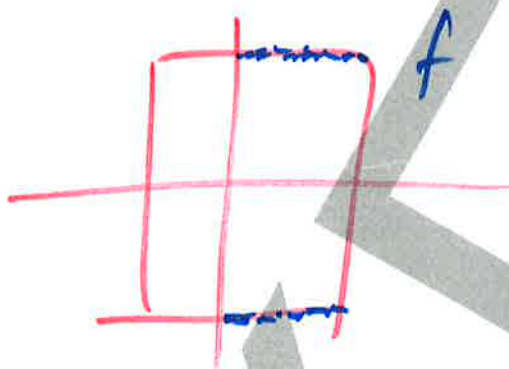
$\rightarrow$  no sequence of steps  
that converges uniformly  
to  $f$ .  $\square$



3. [points] **Prove or disprove** the following statement: Let  $I = [0, 1]$  and  $f$  be a real valued function defined on  $I$ . If  $|f| : I \rightarrow \mathbb{R}$  defined by  $x \mapsto |f(x)|$  is Riemann integrable, then so is  $f$ .

$$f: I \rightarrow \mathbb{R}$$

$$x \mapsto \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ -1, & \text{if } x \notin \mathbb{Q} \end{cases}$$



$$\rightarrow |f| = 1 \Rightarrow \text{int}$$

but we claim that  $f$  itself is not R-int.

$\bar{\sigma} : I \rightarrow \mathbb{R}$  be step,  $\bar{\sigma} \geq f$ .

Then by density of rationals,  $\bar{\sigma} \geq 1$  ✓

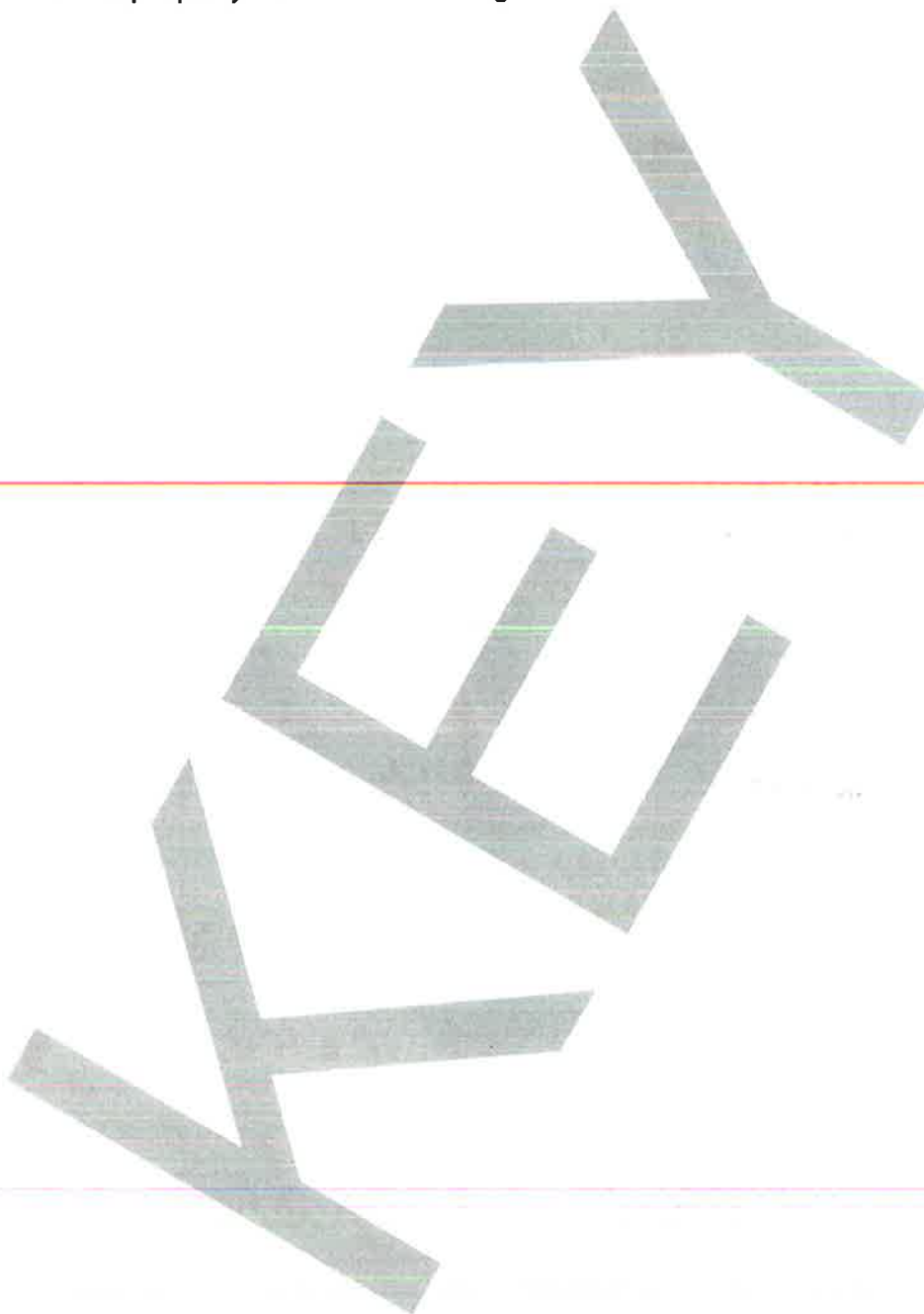
argued  
done  
in class.

$$\int_I \bar{\sigma} \geq 1 \Rightarrow \boxed{\int_I f \geq 1}$$

Similarly if  $\underline{\sigma} : I \rightarrow \mathbb{R}$  is step,  $\underline{\sigma} \leq f$ ,

$$\Rightarrow \int_I \underline{\sigma} \leq -1 \Rightarrow \boxed{\int_I f \leq -1}$$

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4. [4 points] Let  $f, g : \mathbb{R} \rightarrow \mathbb{R}$  be two functions. If both  $f$  and  $g$  are differentiable at 0 and  $g(0) \neq 0$ , then

(a) [2 points] **Prove or disprove the following statement:**

There is an open interval  $I$  containing 0 such that on it the function  $f/g : x \mapsto f(x)/g(x)$  is well defined.

as a real valued function.

$f, g$  diff at 0  $\Rightarrow f, g$  cont. at 0

$g(0) \neq 0 \Rightarrow \exists \delta, \forall x \in \mathbb{R} :$

$|x| < \delta \Rightarrow |g(x)| > 0$  by continuity of  $g$ .

Put  $I = ]-\delta, \delta[$  B.



- (b) [2 points] Prove or disprove the following statement:  
 $f/g$  is differentiable at 0 and

$$(f/g)'(0) = \frac{f'(0)g(0) - f(0)g'(0)}{g(0)^2}.$$

~~Wlog~~ Wlog,  $g(0) > 0$ . Define  $\varphi: \mathbb{R}_{>0} \rightarrow \mathbb{R}$   
 $x \mapsto \frac{1}{x}$   
 Then  $\varphi'(x) = -\frac{1}{x^2}$ ,  $\varphi$  is diff.

product rule (done in class) Then  $\left(\frac{f}{g}\right)'(0) = (f \cdot \varphi \circ g)'(0)$   
 $= f'(0) \cdot \varphi \circ g(0) + f(0) \cdot (\varphi \circ g)'(0)$

$$= f'(0) \cdot \varphi \circ g(0) + f(0) \cdot \varphi' \circ g(0) \cdot g'(0)$$

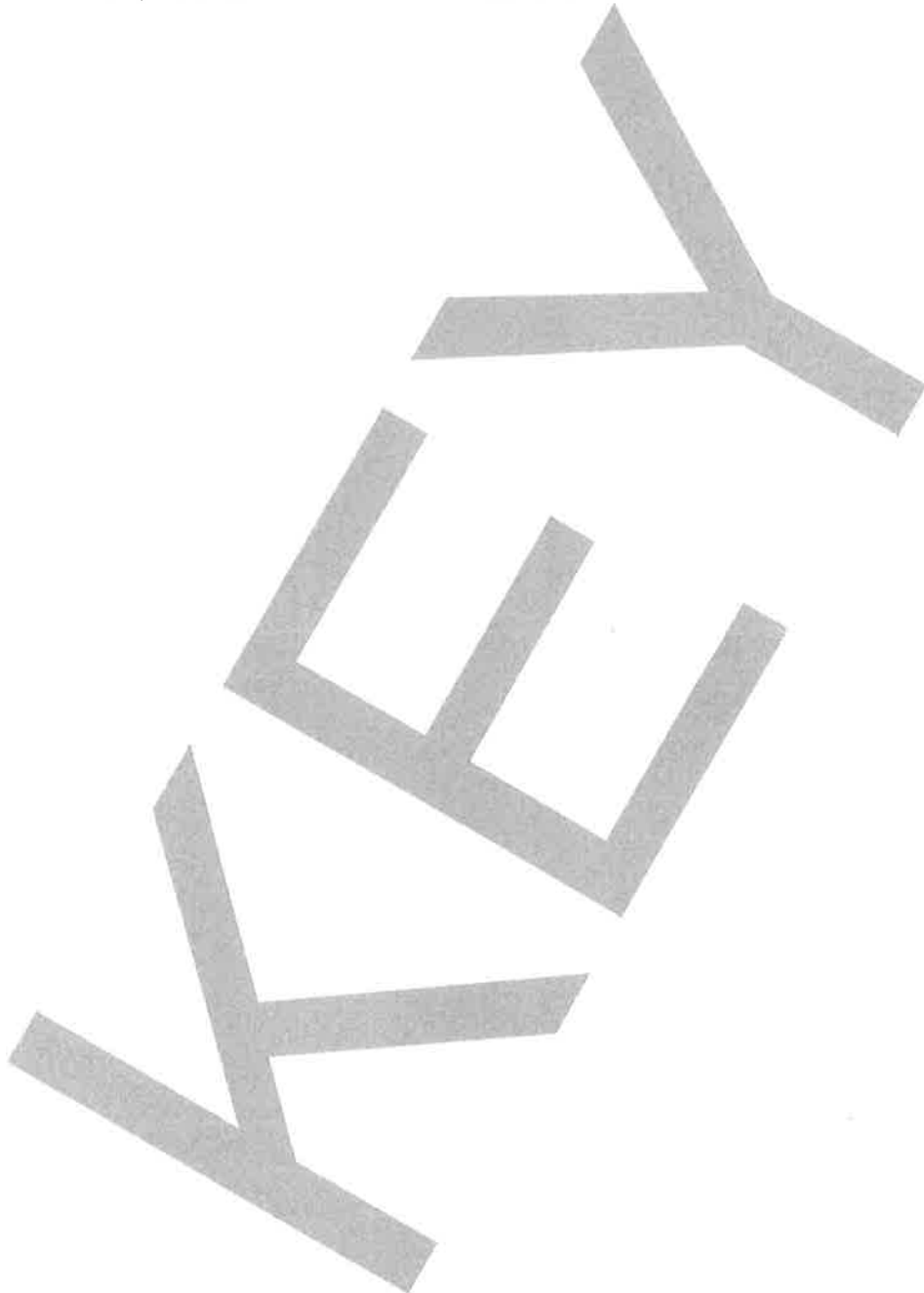
Chain Rule =  $\frac{f'(0)}{g(0)} + f(0) \cdot \frac{-1}{g(0)^2} \cdot g'(0)$   
 (done in class)

$$= \frac{f'(0)g(0) - f(0)g'(0)}{g(0)^2}$$

$g(0)^2$

□.

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